

RCASE Boundary: Public Evidence-Bundle Composition

Gio Della-Libera

Draft — June 28, 2026

Abstract

Public election evidence often arrives as several bounded packages: count reconciliation, unit lineage, audit transcripts, custody records, and public citations. RCASE is reserved for the later layer that can put those materials into one citable case without changing what any child package proves. This paper does not describe implemented code. It records the ownership definition, the reason RCASE is deferred until child claim boundaries are stable, and the rule that a case bundle may organize, cite, and summarize child claims only while preserving child hashes, caveats, conflicts, and unsupported checks.

1 Introduction

A public election case should be able to cite more than one evidence package. Counts may live in RCOUNT, unit lineage in RHIST, audit transcripts in a later audit package, and public exhibits in source citations or custody records. A reader still needs one place to ask which child package supports which public claim.

RCASE is the reserved place for that public bundle. Its purpose is composition, not fact creation. It can order exhibits, point to child package hashes, and summarize what those packages say. It must not turn a count reconciliation into a custody finding, turn a lineage crosswalk into an election total, or make a weak child claim look stronger by placing it in a polished report.

This paper covers the X.4 reservation. It does not claim that an RCASE crate, verifier, schema, fixture, or command exists. It records the intended ownership boundary so the earlier packages can finish their own claim surfaces without accidentally absorbing the public-case role.

2 Ownership Contract

RCASE owns child-package composition and a claim matrix. A case bundle is allowed to say, for each public claim, which child packages are being cited, what their hashes are, and which child verifier results, caveats, conflicts, or unsupported checks remain attached to the

claim. The planned shapes are `claims/claim-boundary.json` and a child-package index, but those are reserved interfaces rather than implemented schemas.

RCASE may	RCASE must not
Organize child packages into a public evidence bundle.	Create facts that are not available in the cited children.
Cite child package hashes and external citations.	Replace child hashes with a case-level assertion that cannot be traced back.
Summarize child package claims in a claim matrix.	Loosen caveats, hide unsupported checks, or make an inconclusive child result look conclusive.
Preserve conflicts between child packages.	Merge conflicting claims without recording the conflict metadata.
Order exhibits for a public reader.	Move ownership of counts, lineage, audits, custody, maps, or certification into the case layer.

The contract is intentionally narrow. RCASE can make a case easier to read and easier to cite, but the evidence remains owned by the child packages that declared or derived it.

3 Deferral and Boundary

RCASE is deliberately deferred. The downstream-boundary specification defines RCASE as the owner of public evidence bundles and claim matrices, while the layer-access specification places it after the child packages whose hashes and claim boundaries it reads. That order matters: a composition package cannot state its own schema honestly until the child packages have stable ways to say what they prove, what they do not prove, and what their verifiers could not check.

The first implementation step should therefore not be a broad report generator. It should be a tiny composition fixture after RCOUNT and RHIST claim boundaries are stable. That fixture can exercise the child-package index and `claims/claim-boundary.json` without pretending that RCASE owns count arithmetic, unit lineage, or audit method replay.

The boundary rule is the useful part now: RCASE may organize, cite, and summarize child package claims. It must preserve child hashes, caveats, conflicts, and unsupported checks. If a child package reports a failed verifier, a missing source, an unresolved conflict, or a scoped claim, the case layer must carry that condition forward instead of laundering it into a clean public sentence.

4 Conclusion

RCASE is a reservation for public composition. It gives the civic evidence family a future place to assemble child packages into one citable case, but it does not create a new source of election facts. The case layer can improve navigation, citation, and public readability only by preserving the evidence boundaries below it.

Keeping RCASE deferred is part of that design. Once RCOUNT, RHIST, and related children have stable claim-boundary records, a small RCASE fixture can test the composition surface. Until then, the enforceable rule is simpler: no new facts, no hidden caveats, no erased conflicts, and no case-level claim that cannot be traced back to a child package hash.

References

Gio Della-Libera. Civic evidence layer access patterns and downstream boundaries. Technical report, Apportionment Research, 2026a.

Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026b.